

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 17+ years of experience building production applications in collaborative engineering environments. Offers strong evidence in OpenAI, AI Agents, AI-Assisted Development, Testing, Azure Storage Queues, and Containerization, with additional hands-on background in Codex, Python, and JavaScript. Experience spans APIs, data-intensive systems, automation platforms, and cloud-based application delivery.

Selected Skills

Testing, AI Agents, AI-Assisted Development, Azure Storage Queues, Containerization, OpenAI, System Architecture, Tailwind CSS, .NET, Application Modernization, ASP.NET, Auth0, Authentication, Azure, Azure App Service, Azure DevOps, Azure Functions, Azure SQL, Bootstrap, Business Process Automation, C#, CI/CD, Code Review, Codex

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Implemented AI bots integrated with Microsoft Teams.
- Designed and implemented tens of Dataverse tables to support the application's data model.
- Created dozens of Power Automate flows and dataflows to support compliance workflows and data movement.
- Developed a React application embedded inside a model-driven Power Apps application.
- Built a complex User Access Review compliance system on the Power Platform for NBCUniversal.

Enerex - AI Development Consultant

2023-11 - present | Remote - New Jersey

Architected frontend and AI service layers for task-based automation systems and cross-product integrations.

- Provisioned and deployed Azure Functions, web applications, databases, and Storage Queues to support Enerex products.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users.
- Designed the add-in to open directly from contract emails, analyze attached contracts with an agent carrying deep business knowledge, and return standardized extracted data.
- Built production applications with Codex as a primary development tool, accelerating implementation across frontend, AI service, and delivery work.
- Implemented deployment pipelines for application delivery and environment updates.

First Due - Senior Software Engineer

2017-07 - 2019-11 | Remote

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Improved control and accountability of company web-crawling operations through centralized management and visibility.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Architected a system to manage and control web crawlers, centralizing execution control and operational monitoring in a single UI.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06 | Remote

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Implemented and ran SQL queries to investigate, validate, and repair data issues.
- Implemented numerous stored procedures to support application behavior and database operations.
- Migrated projects from Visual Basic to ASP.NET as part of legacy application modernization work.
- Performed extensive database data consistency reviews and corrections across business-critical datasets.

DATYS - Computer Science Specialist

2009-09 - 2016-04

Worked on FPGA research, automation tooling, and distributed systems for fraud detection and identification platforms.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Integrated Nylas and Google services to enable controlled access to user email inboxes for invoice ingestion.
- Implemented AI-powered invoice analysis using OpenAI.
- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | Havana, Cuba | 2004 - 2009